KUSH MEHTA

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Electronics & Communication Engineering student (Minor: Data Science) who builds things end-to-end — from a real-time C++ decoder shipped during a research internship at ISRO, to encrypted IoT hardware, computer vision, and data pipelines. Three years leading the electronics team for competition UAVs, and a co-authored paper under review. Looking for engineering internships in software, embedded systems, or data.

EDUCATION

Institute of Technology, Nirma University — Ahmedabad, India
B.Tech, Electronics & Communication Engineering — Minor: Data Science | CGPA: 8.59 / 10 — Aug 2023 – May 2027

- **Coursework:** Computer Programming, Embedded Systems, Computer Architecture, Microcontrollers & Interfacing, Scripting Languages, Programming for Scientific Computing, VLSI Design, Digital Signal Processing, Communication Systems · *Data Science minor:* Machine Learning, Statistics, Data Analysis & Visualization, Analytics of IoT

EXPERIENCE

Research Intern — Navigation Receiver Division, Space Applications Centre (SAC), ISRO — May – Jul 2026
Real-Time Galileo High Accuracy Service (HAS) Decoding Utility · 10-week research internship — Ahmedabad, India

- Built a real-time C++ utility that ingests a multiplexed TCP stream, tokenizes three interleaved protocols (RTCM payloads, Galileo HAS E6-B pages, Time-of-Transfer epoch tags), and routes each to independent output ports.
- Implemented Reed–Solomon (255, 32) erasure decoding over GF(2^8) from the Galileo HAS ICD — lookup-table field arithmetic and Gaussian elimination for k×k matrix inversion — recovering complete messages despite page loss and out-of-order arrival.
- Reverse-engineered four undocumented Galileo RTCM 3.x message formats (1240–1243) and the CRC-24Q framing routine by cross-referencing an open-source implementation against the ICD.
- Delivered real-time streaming over TCP and serial with background-thread reads and stale-buffer purging, working around a pre-C++11 toolchain that ruled out modern STL containers and threading.
- Validated end-to-end in RTKLIB at centimetre-level positioning, with each decoding stage unit-verified against a Python reference implementation.

Electronics Lead — Team Arrow UAV Club, Nirma University — Sep 2023 – Present
Official Aeromodelling & Autonomous Systems Club — Ahmedabad, India

- Own the avionics and embedded stack for competition aircraft: flight-controller configuration (ArduPilot / Pixhawk), telemetry links, sensor integration, ESC and motor selection, and power system design.
- As propulsion lead for the 2025 international entry, engineered the power plant under a hard 400 W competition limit — motor and propeller matching validated on a thrust bench against analytical predictions.
- Lead a team of 8–12 and mentor juniors on firmware, sensor fusion, and avionics integration. Four podium finishes across national and international SAE and Technoxian competitions, including AIR 2 (SAE DDC Micro 2024) and 5th internationally in mission score (SAE Aero Design West 2025, USA).

PROJECTS

Autonomous Tilt-Rotor VTOL Tricopter with AGV Payload — SAE Aero Design East 2026 — 2025 – 2026
Electronics & Avionics Lead · Team Arrow, Nirma University

- Led the avionics stack for a hybrid tilt-rotor tricopter that takes off vertically, transitions to cruise, and flies autonomously: TBS Lucid H7 controller with Here4 GPS for centimetre-level navigation, a servo-actuated tilt mechanism with airspeed-dependent interlock to prevent stall during transition, and power design to 2:1 thrust-to-weight.
- Designed the Autonomous Ground Vehicle payload electronics — magnetometer, time-of-flight ranging, IR line-following — so the vehicle self-navigates the target zone, qualifying for the highest 14× capture multiplier.
- Validated the full mission in software-in-the-loop before first flight; my scoring analysis drove the team's core strategy — fully autonomous low-payload scores ~52 points against ~30 for manual high-payload.

Secure GPS Telemetry Pipeline — ESP32, AES-128, MQTT — Mar 2026

- End-to-end encrypted IoT pipeline: ESP32 parses NMEA over UART2 with TinyGPS++, serializes to JSON, encrypts with AES-128 via mbedTLS, and publishes hex ciphertext to an MQTT broker; a Python subscriber (paho-mqtt, PyCryptodome) decrypts and resolves the live position. Verified on hardware end to end.

Real-Time Virtual Try-On — Computer Vision, OpenCV — Sep 2025

- Real-time AR try-on built entirely with classical computer vision — no deep learning or GPU: CLAHE contrast enhancement, Haar Cascade face and eye detection, pose estimation from inter-pupillary distance, and per-pixel alpha compositing. An exponential moving average (α=0.2) on position, scale, and rotation removed jitter without lag.

Cryptography Benchmark for Resource-Constrained Devices — Jan 2026

- Benchmarked AES-GCM, DES, ECC P-256, and ASCON-128 across 5,000 IoT payloads; loaded the ASCON C reference as a compiled DLL through Python `ctypes` for native-speed timing, profiling peak memory with `tracemalloc`. ASCON ran 78× faster than AES-GCM and 12× faster than DES.

PUBLICATIONS & PATENTS

- M. Bhavsar, **K. Mehta**, P. Parikh, S. Gajjar, "Jamming Utilization: An Evaluation Hazard in Learned Frequency-Hopping Receivers." *Under review, 2026.* Showed a learned hop detector scores higher under attack (0.842) than in the clear (0.712) while retaining no genuine detection ability (1.8% against 1.6% chance), and proposed the control metric that exposes it.
- **Patent application — filing in progress, 2026.** Micro-class aircraft design developed for SAE India DDC 2025. Procedure initiated; not yet granted.

TECHNICAL SKILLS

Languages: Python, C, C++, Verilog HDL, Bash, SQL, MATLAB, Assembly, TCL
Software & Data: pandas, NumPy, Matplotlib, scikit-learn, PyTorch, OpenCV, KNIME, Tableau, object-oriented design, Git / GitHub
Embedded & Hardware: ESP32, Raspberry Pi, Arduino, 8051, Pixhawk flight controllers, ArduPilot / Mission Planner, Keil uVision, mbedTLS, UART / I²C / SPI, FPGA
Systems & Networking: Linux / Ubuntu, TCP/IP sockets (WinSock), Win32 API, multithreading, cron, MQTT (HiveMQ, Mosquitto), Wi-Fi, BLE
Simulation & Tools: gem5, ModelSim, Quartus II, RTKLIB, SolidWorks, Fusion 360